

Discovery of Aficamten (CK-274), a Next-Generation Cardiac Myosin Inhibitor for the Treatment of Hypertrophic Cardiomyopathy

Chihyuan Chuang,* Scott Collibee, Luke Ashcraft, Wenyue Wang, Mark Vander Wal, Xiaolin Wang, Darren T. Hwee, Yangsong Wu, Jingying Wang, Eva R. Chin, Peadar Cremin, Jeanelle Zamora, James Hartman, Julia Schaletzky, Eddie Wehri, Laura A. Robertson, Fady I. Malik, and Bradley P. Morgan*

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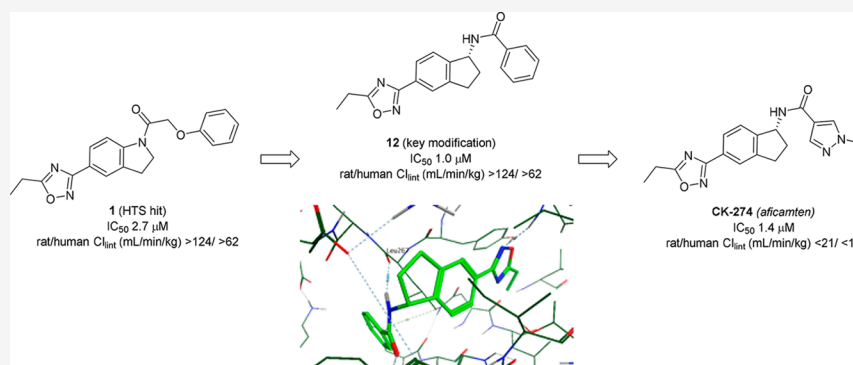
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ABSTRACT: Hypercontractility of the cardiac sarcomere may be essential for the underlying pathological hypertrophy and fibrosis in genetic hypertrophic cardiomyopathies. Aficamten (CK-274) is a novel cardiac myosin inhibitor that was discovered from the optimization of indoline compound **1**. The important advancement of the optimization was discovery of an Indane analogue (**12**) with a less restrictive structure–activity relationship that allowed for the rapid improvement of drug-like properties. Aficamten was designed to provide a predicted human half-life ($t_{1/2}$) appropriate for once a day (qd) dosing, to reach steady state within two weeks, to have no substantial cytochrome P450 induction or inhibition, and to have a wide therapeutic window *in vivo* with a clear pharmacokinetic/pharmacodynamic relationship. In a phase I clinical trial, aficamten demonstrated a human $t_{1/2}$ similar to predictions and was able to reach steady state concentration within the desired two-week window.

INTRODUCTION

Hypertrophic cardiomyopathy (HCM) is an inherited cardiovascular disorder normally characterized by abnormal thickening of the left ventricular walls that consequently reduces stroke volume and limits cardiac output.¹ HCM is one of the most common classes of genetic disease in young adults. It is estimated that 1 in 500 people carry genetic mutations that increase susceptibility to HCM,² with approximately 1 in 3200 developing HCM.^{3–5} About two-thirds of HCM patients develop obstructive HCM (oHCM), where asymmetric septal hypertrophy causes outflow obstruction of blood leaving the left ventricle and results in increases of the left ventricular outflow tract (LVOT) pressure gradient.⁶ Current treatment for HCM is limited to symptomatic relief and does not treat the root cause of the disease, excessive sarcomere contractility. Pharmacological treatments include beta blockers (e.g., metoprolol, propranolol), calcium channel blockers (e.g., verapamil, diltiazem), antiarrhythmic drugs (e.g., amiodarone, disopyramide), and anticoagulants to prevent thrombus

formation. Surgical procedures to treat symptomatic HCM include heart muscle reduction procedures (septal myectomy surgery and alcohol septal ablation), defibrillator placement, and heart transplant for the most refractory cases.

Hypertrophic cardiomyopathy, whose earliest indicators are diastolic dysfunction with either preserved or hypercontractile systolic function, generally involves mutations of the proteins within the cardiac sarcomere.¹ A leading hypothesis currently under investigation is that “HCM mutation–induced hypercontractility leads to activation of signaling pathways that cause hypertrophy of the heart, fibrosis, and myofilament disarray”.⁷ Direct inhibition of cardiac sarcomere contractility may be a

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viable approach to treat patients with HCM by restoring the proper degree of contractility within the sarcomere.⁸ Mavacamten was the first small molecule allosteric inhibitor of cardiac myosin, the mechanochemical enzyme that drives muscle contractility, to advance into clinical trials (Figure 1).⁹

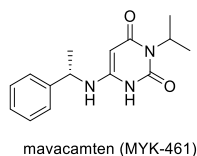


Figure 1. Structure of mavacamten.

In a phase 3 trial in patients with oHCM using a pharmacokinetic and echo-based dose-titration strategy to find the optimal dose for each patient, mavacamten demonstrated improvements in peak oxygen consumption (pV_{O_2}) and NYHA (New York Health Association) class as well as a reduction in the LVOT pressure gradient, thus demonstrating a clinical benefit of this therapeutic approach in oHCM patients.¹⁰ The long human half-life ($t_{1/2}$) of mavacamten, approximately 7–9 days, requires about a 6-week period to reach steady state concentrations,^{11–13} a significant amount of time for each patient to reach the targeted dose using an individualized dose titration strategy, and also a long wash-out period in the event of excessive on-target pharmacology. Additionally, mavacamten induced CYP3A4 and CYP2B6 in plated human hepatocyte studies (EC_{50} of 2.2 ± 0.4 and 5.1 ± 0.2 μ M, respectively),¹¹ suggesting a potential for drug–drug interactions (DDIs) dependent on the clearance mechanisms of concomitant medications.

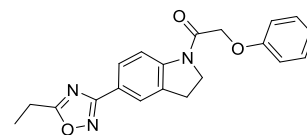
To maximize the safety, efficacy, and ease of use for the patient, the design of a next-generation cardiac myosin inhibitor (CMI) focused on three primary objectives. The first objective was to design a compound with a human $t_{1/2}$ appropriate to allow once a day (qd) dosing and to reach steady state plasma concentrations within two weeks, minimizing the peak to trough ratio which is an important consideration for a cardiac drug and facilitating the ability of clinicians to safely titrate the dose to reach therapeutic concentrations. Second, the next-generation CMI should have as wide a therapeutic window as possible with a clear pharmacokinetic (PK)/pharmacodynamic (PD) relationship that can be measured efficiently in preclinical experiments. The third primary objective was the design of a compound with no substantial cytochrome (CYP) P450 induction or inhibition to minimize the potential for CYP-induced drug–drug interactions.

RESULTS AND DISCUSSION

A high-throughput screen (HTS) using bovine cardiac muscle myofibrils measuring the rate of ATP hydrolysis was performed, and several novel chemical structures were identified as new starting points for optimization.¹⁴ Among the HTS hits, compound 1 had several desirable features for a primary hit (Table 1). Indoline 1 displayed single digit micromolar potency against cardiac myofibrils and showed selectivity with no measurable inhibition against smooth muscle myosin. Exquisite selectivity for cardiac vs smooth muscle activity was observed throughout the optimization of

Table 1. Properties of HTS Hit Compound 1

property	data
cardiac myofibril IC_{50} (μ M)	2.7
smooth muscle myosin (SMM) IC_{50} (μ M)	>39
ligand efficiency	0.28
cLogP	3.5
rat microsome, Cl_{int} , mL/min/kg	>124
human microsome, Cl_{int} , mL/min/kg	>62



this series and was not a necessary optimization parameter. Additionally, 1 inhibited contractility in primary rat cardiomyocytes without perturbation of the calcium transient as discussed below. Compound 1 was rapidly oxidized in both human and rat microsomes, necessitating optimization of this parameter.

The effect of compound 1 on the contractility of a rat cardiac myocyte is shown in Figure 2A. A significant reduction

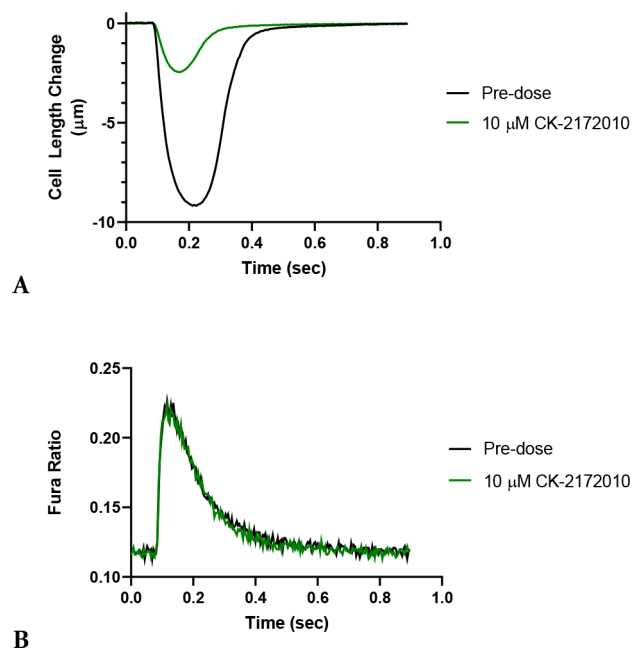
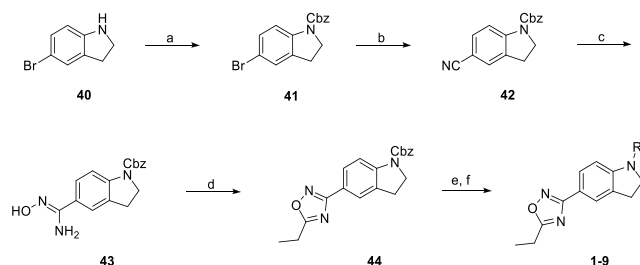


Figure 2. Myocyte activity of screening hit 1. (A) Cardiomyocyte contractility profile of compound 1 (10 μ M) vs vehicle. (B) Effect on calcium concentration in cardiomyocytes of compound 1 (10 μ M) vs vehicle determined by a Fura dye.

in myocyte contractility as measured by myocyte length was observed upon treatment with 1 (10 μ M) relative to a predose measurement. This result is consistent with the reduced ATPase rate observed in the biochemical assay. The reduction of contractility was achieved without changing calcium concentrations within the myocyte and provided evidence that inhibition of contractility was occurring via direct interaction of 1 with the cardiac sarcomere (Figure 2B).

The synthetic route for analogues 2–9 to enable exploration of the amide substituent of the indoline core is described in Scheme 1. Synthesis of the indoline intermediate was achieved in five steps starting with benzyloxycarbonyl (Cbz) protection

Scheme 1. Synthesis of Indoline Analogues 2–9^a

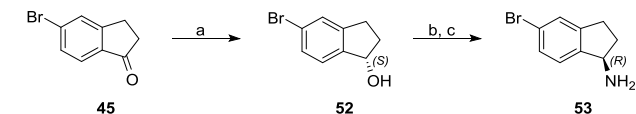
^aStandard conditions: (a) CbzCl, pyridine. (b) (Ph₃P)₄Pd, Zn(CN)₂, DMF. (c) H₂NOH, EtOH. (d) ClCOCH₂CH₃, pyridine. (e) Pd/C, H₂, MeOH, CH₂Cl₂. (f) PhCH₂COH, NaB(OAc)₃H, DCE or RCOCl, pyridine or RNCO, DMAP, THF.

of commercially available 5-bromoindoline (40), followed by a palladium-mediated cyanation to provide nitrile 42. Oxadiazole 44 was generated by treatment of nitrile 42 with hydroxylamine to yield the hydroxycarbamidoyl intermediate 43 followed by cyclization with propionyl chloride in the presence of pyridine.¹⁵ Removal of the Cbz group gave the indoline intermediate that was then acylated to produce 2–8 or alkylated to give 9. Analogues 10 and 11 were synthesized using an analogous route starting with 6-bromoindoline and 7-bromoindoline, respectively.

The general synthesis of 2,3-dihydro-1*H*-indene analogues is shown in Scheme 2. Alternate synthetic routes were designed to enable rapid SAR exploration of both the oxadiazole substituent and the amide position. Reductive amination of 5-bromo-1-indanone (45) using ammonium formate and sodium cyanoborohydride provided amine 46, which was acylated using either acid chlorides or carboxylic acids in the presence of an appropriate coupling agent to give the corresponding amides (47). Cyanation of the resultant aryl bromide was performed using potassium ferrocyanide hydrate, and the oxadiazole ring was incorporated using the same two-step sequence described in Scheme 1. Alternatively, amine 46 was protected with a *tert*-butoxycarbonyl (Boc) group and the resultant bromide (49) was converted to an oxadiazole using the same three step sequence previously described. Deprotection of Boc amide 51 provided an intermediate amine used to make amide analogues 12–20.

Chiral cardiac sarcomere inhibitors were synthesized using (R)-5-bromo-2,3-dihydro-1*H*-inden-1-amine (53). The three

step, asymmetric synthesis of intermediate 53 starting with 5-bromo-1-indanone (45) is shown in Scheme 3. Chiral alcohol

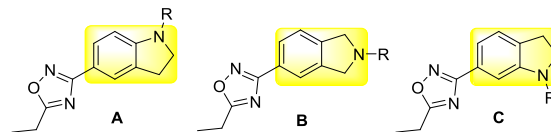
Scheme 3. Asymmetric Synthesis of (R)-5-Bromo-2,3-dihydro-1*H*-inden-1-amine^a

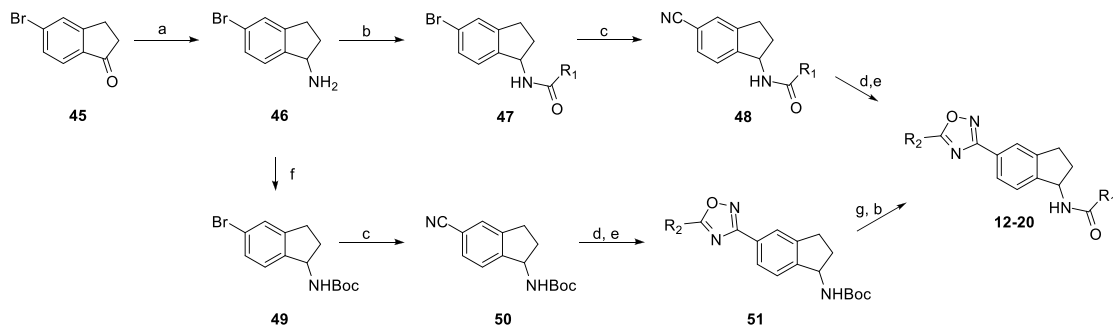
^aStandard conditions: (a) (R)-(-)-2-methyl-CBS-oxazaborolidine, BH₃·Me₂S, THF. (b) DPPA, DBU, toluene. (c) SnCl₄·2 H₂O, MeOH.

52 was synthesized using a Corey–Bakshi–Shibata (CBS) asymmetric reduction of ketone 45.¹⁶ The resultant alcohol was then converted to an azide by treatment with diphenylphosphoryl azide, followed by reduction with tin chloride hydrate to give chiral amine intermediate 53.¹⁷

In the absence of structural information on the binding site for compound 1 on cardiac myosin, initial optimization sought to explore the amide moiety and spatial relationship of the substituents on the indoline core. The inhibition of bovine cardiac myofibrils from this initial exploration is shown in Table 2 and is expressed as both the half maximal inhibitory

Table 2. Indoline Analogues

					
#	core	R	IC ₅₀ (μM)	IC ₁₅ (μM)	
1	A	COCH ₂ OPh	2.7	0.9	
2	A	CO ₂ CH ₂ Ph	>39	1.3	
3	A	COPh	>39	>39	
4	A	COCH ₂ Ph	19.1	3.5	
5	A	COCH ₂ CH ₂ Ph	>39	>39	
6	A	COCH ₂ SPh	>39	>39	
7	A	CONHPh	>39	>39	
8	A	CONHCH ₂ Ph	>39	>39	
9	A	CH ₂ CH ₂ Ph	>39	3.4	
10	B	COCH ₂ Ph	>39	15.9	
11	C	COCH ₂ Ph	>39	>39	

Scheme 2. General Synthesis of 1,2,4-Oxadiazol-3-yl-2,3-dihydro-1*H*-inden-1-amine Analogues^a

^aStandard conditions: (a) HCO₂NH₄, MeOH, NaBH₃CN. (b) RCOCl, TEA, CH₂Cl₂ or R₁COOH, EDC, HOBT, DIEA, DMF. (c) K₄Fe(CN)₆·3H₂O, XPhos Pd G2, KOAc, dioxane/water. (d) H₂NOH·HCl, TEA, EtOH. (e) (RCO₂)₂O, dioxane or ClCOR, pyridine. (f) Boc₂O, TEA, CH₂Cl₂. (g) TFA, CH₂Cl₂.

concentration (IC_{50}) and the inhibitor concentration needed to inhibit maximal activity by 15% (IC_{15}). The IC_{15} values provided a greater activity window to enable a more granular perception of the structure–activity relationship (SAR) and were thought to be more correlative to the expected maximal efficacious free concentrations needed *in vivo*.¹⁸ Small changes in the amide group of **1** such as moving the position of the oxygen (carbamate **2**) and replacement of the ether oxygen with a carbon (amide **5**) or sulfur (thioether **6**) led to a reduction in biochemical potency. Phenyl acetamide **4** was the only new amide with a measurable IC_{50} value at 19.1 μ M. The spatial preference of the 5-oxadiazole isomer was demonstrated by the decreased activity in positional isomers **10** and **11** as compared to **4**.

At this point in the SAR exploration, compounds within the chemical series were shown to bind to cardiac myosin and exclude the known myosin inhibitor blebbistatin from binding to cardiac myosin; implying that these two compounds could potentially share a similar binding site.^{19,20} The crystal structure of blebbistatin bound to an analogous myosin II from *Dictyostelium discoideum* shows a key hydrogen bond between the angular hydroxyl moiety of blebbistatin and a backbone carbonyl (Leu 262) of myosin II (Figure 3A).^{21,22} It was hypothesized that a hydrogen bond donor may be added to the core of our analogues by replacing the indoline core of **4** (Table 2) with the 2,3-dihydro-1H-inden-1-amine core of **12** (Table 3) and enable a similar H-bond between **12** and a backbone carbonyl (Leu 267) in cardiac myosin as shown in Figure 3B. This model also suggested that the phenyl ring from the benzamide of **12** may fit into a similar hydrophobic pocket as the *N*-phenyl from the pyrrolidine ring of blebbistatin. According to this model, the oxadiazole of **12** may be able to form an H-bond with a backbone N–H from alanine 463 in cardiac myosin and allow for the ethyl substituent of **12** to occupy a similar space as the methyl substituent of blebbistatin. Compound **12** displayed an approximately 20 $\times$ potency improvement compared to **4** (IC_{50} values of 1.0 and 19.1 μ M, respectively) and provided a new core to optimize drug-like properties.

The initial optimization of 2,3-dihydro-1H-inden-1-amine **12** focused toward compounds with reduced lipophilicity to decrease intrinsic clearance in human microsomes and improve drug-like properties. A series of racemic amide analogues were prepared to rapidly assess the SAR of this novel core (Table 3). Truncation of the oxadiazole ethyl substituent of **12** to a methyl group (**13**) improved human microsome stability (CL_{int} > 62 and 24 mL/min/kg, respectively) and increased rat plasma protein free fraction (2 and 4%, respectively) with only a 2-fold reduction in biochemical potency. In combination with this modest improvement in drug-like properties, the SAR exploration focused on replacement of the benzoyl group of **13** with less lipophilic (according to calculated log *P* values) heterocyclic amides (**14**–**20**) to reduce turnover in microsomes and increase free drug exposure. Although there was a modest attenuation of biochemical potency for all analogues relative to **13**, rat and human microsome stability was significantly improved along with greater free fraction in rat plasma protein.

To assess the stereochemical preference for biological activity, 1-methyl-*N*-(5-(5-methyl-1,2,4-oxadiazol-3-yl)-2,3-dihydro-1H-inden-1-yl)-1H-pyrazole-4-carboxamide was separated using chiral chromatography into its *R* and *S* enantiomers (**17** and **18**, Table 4). Stereochemical assignments were

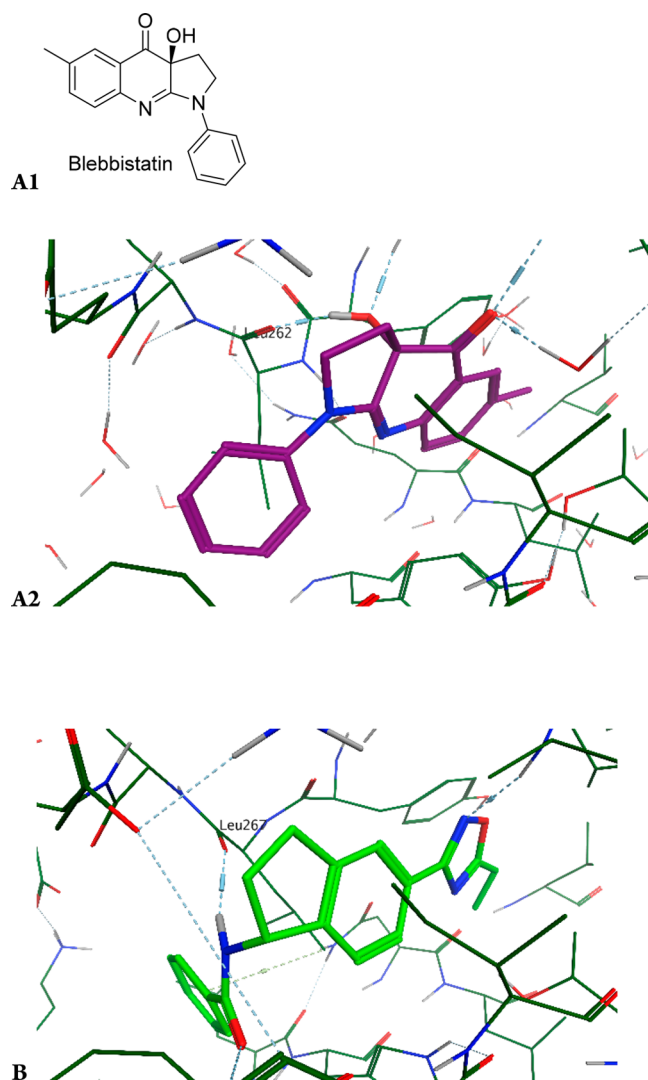
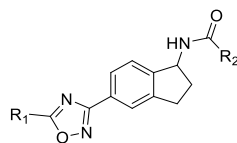


Figure 3. (A1) Chemical structure for blebbistatin. (A2) Representation of the binding pocket of blebbistatin (purple) with myosin II from *Dictyostelium discoideum* from the X-ray coordinates.²¹ (B) Molecular model of compound **12** (*R*) docked into a similar binding pocket with cardiac myosin.

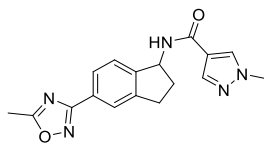
confirmed through asymmetric synthesis of the *R*-enantiomer (compound **17**). A greater than 8-fold eudismic ratio was observed with compounds **17** and **18** in the bovine cardiac myofibril ATPase assay, and the *S*-isomer (**18**) showed an IC_{50} value greater than 39 μ M. Furthermore, the evaluation of clearance (CL) in rats revealed that there is a difference of clearance values observed in liver microsomes vs the *in vivo* rat system (Table 4) for this pair of enantiomers. The *R*-isomer demonstrated a much lower CL (2.1 mL/min/kg) and free drug CL (32.3 mL/min/kg) than the *S*-isomer (57.7 mL/min/kg for CL and 395.2 mL/min/kg for free drug CL). The rat *in vitro* microsome clearance of both isomers was <21 mL/min/kg. Subsequent explorations of the 2,3-dihydro-1H-inden-1-amine core were conducted using the *R*-isomer.

With improvements in physical properties and concomitant improvement in microsomal turnover and increases in rat free fraction, the SAR exploration again focused toward improving cardiac myofibril potency by examining the SAR around the methyl substituent of the oxadiazole ring of **17** (Table 5). In general, small lipophilic substituents were most favored at this

Table 3. Assessment of Amino-Substitution (R^2)

#	st ^a	R ¹	R ²	IC ₅₀ (μM)	CL _{int} (r) ^b (mL/min/kg)	CL _{int} (h) ^b (mL/min/kg)	PPB (r) ^c (% free)	cLogP (ACD)
12	rac	Et	Ph	1.0	>124	>62	2.3	3.7
13	rac	Me	Ph	1.9	>124	24	4.3	3.2
14	rac	Me		7.1	<21	<11	9.7	2.4
15	rac	Me		8.1	48	<11	7.2	1.3
16	rac	Me		7.6	<21	<11	8.9	1.7
17	R	Me		4.4	<21	<11	6.5	1.5
18	S	Me		>39	<21	<11	15	1.5
19	rac	Me		5.1	<21	<11	19	0.9
20	rac	Me		2.5	46	<11	6.4	1.3

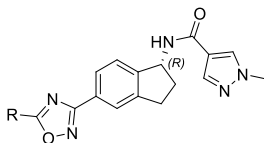
^ast = indane stereochemistry. ^bCL_{int} = intrinsic clearance in rat (r) or human (h) microsomes. ^cPPB = plasma protein binding as % unbound in rat (r).

Table 4. Comparison of *in vitro* Microsome Clearance and *in vivo* Clearance in Rats for 17 and 18

#	st ^a	IC ₅₀ (μM)	CL _{int} (r) ^b (mL/min/kg)	CL (r) ^c (mL/min/kg)	CL _u (r) ^d (mL/min/kg)	PPB (r) ^e (% free)
17	R	4.4	<21	2.1	32.3	6.5
18	S	>39	<21	57.7	395.2	15

^ast = indane stereochemistry. ^bCL_{int} = intrinsic clearance in rat (r) microsomes. ^cCL = *in vivo* rat clearance dosed at 0.5 mg/kg. ^dCL_u = unbound *in vivo* clearance in rat (r) (*in vivo* rat clearance/free fraction in rats). ^ePPB = plasma protein binding, % unbound, rat (r).

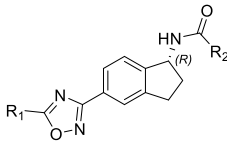
Table 5. Exploration of the 5-Position of the Oxadiazole Ring



#	R	IC ₅₀ (μM)	CL _{int} (r) ^a (mL/min/kg)	CL _{int} (h) ^a (mL/min/kg)	PPB (r) ^b (% free)	PPB (h) ^b (% free)	solubility ^c (μM)	cLogP (ACD)
21	H	>39	-	-	-	-	-	1.0
17	CH ₃	4.4	<21	<11	6.5	14	86	1.5
aficamten	Et	1.4	<21	<11	1.8	9.5	97	2.1
22	cPr	2.7	-	-	4.2	5.6	33	1.9
23	iPr	4.1	-	-	-	-	-	2.4
24	CHF ₂	2.4	<21	<11	2.4	5.7	37	2.0
25	cBu	3.2	-	-	-	-	14	2.5
26	CH ₂ OMe	6.8	<21	<11	8.2	12	63	1.4
27	CH ₂ CH ₂ OMe	>39	-	-	-	-	-	1.5
28	CH ₂ CH ₂ OH	>39	-	-	-	-	-	0.9

^aCL_{int} = intrinsic clearance in rat (r) or human (h) microsomes. ^bPPB = plasma protein binding, % unbound, rat (r) or human (h). ^cSolubility = shake flask solubility at pH 6.8, solid form was not characterized. "-" = not tested.

Table 6. Final Optimization



#	R ₁	R ₂	IC ₅₀ (μM)	Rat CL ^a (mL/min/kg)	Rat t _{1/2} ^{ba} (h)	Rat F ^c (%)	CL _{int} (r) ^d (mL/min/kg)	CL _{int} (h) ^d (mL/min/kg)	PPB (r) ^e (% free)	PPB (h) ^e (% free)
aficamten	Et		1.4	2	3.0	55-79	<21	<11	1.8	9.5
29	Et		1.7	21	4.1	13	<21	<11	4.6	18
30	Et		0.9	20	6.3	53-60	<21	<11	11	29
31	Et		0.8	40	14.7	0	>124	<11	5.1	10
32	Et		1.1	63	1.4	10	66	<11	20	20
33	Et		1.2	13	3.5	32	23	<11	2.1	8.4
34	Et		1.2	44	6.6	16	-	<11	-	31
35	Et		1.8	20	5.9	26	24	<11	4.6	9.2
36	cPr		1.0	5	17.8	100	<21	<11	7.7	12
37	iPr		1.9	16	4.6	52-88	<21	<11	8.9	17
38	cPr		1.9	3	13.8	70	<21	<11	-	4.9
39	iPr		2.7	14	4.6	79	23	<11	-	5.3

^aRat CL = *in vivo* rat clearance dosed at 1 mg/kg. ^bRat t_{1/2} = *in vivo* half-life in rat iv experiment. ^cRat F = *in vivo* bioavailability for compounds dosed iv and po at 1 mg/kg except compound 30, which was dosed po between 2 and 6 mg/kg, compound 36 which was dosed po at 2 mg/kg, compound 37 which was dosed po between 2 and 12 mg/kg, and aficamten which was dosed po between 2 and 8 mg/kg. ^dCL_{int} = intrinsic clearance in rat (r) or human (h) microsomes. ^ePPB = plasma protein binding, % unbound, rat (r) or human (h). “-” = not tested.

Table 7. Animal PK and Human PK Projections for Aficamten

species ^a	dose level (mg/kg)	route	C _{max} ^b (ng/mL)	t _{max} ^c (h)	AUC _{0-t} ^d (ng·h/mL)	CL ^e (mL/min/kg)	V _{ss} ^f (L/kg)	t _{1/2} ^g (h)	F (%) ^h
mouse	0.5	iv	-	-	925	8.8	3.14	4.5	-
	1	po	268	0.25	1832	-	-	3.4	98
rat	1	iv	-	-	8564	2.1	0.53	3.0	-
	2	po	1647	0.92	9450	-	-	2.6	55
	3	po	3168	0.42	14 775	-	-	3.5	58
	8	po	8437	0.58	53 711	-	-	3.3	79
dog	1	iv	-	-	2871	3.3	10.5	33.8	-
	1	po	245	0.33	1474	-	-	28.7	45
monkey	1	iv	-	-	1344	11.25	7.76	8.1	-
	1	po	65	1.17	528	-	-	12.1	41
human	-	-	-	-	-	2.1 ⁱ	12.2 ⁱ	68 ⁱ	-

^aAll studies were performed in male animals, *n* = 3 per group, except dog IV (*n* = 2 animals). ^bC_{max} = maximum compound concentration. ^ct_{max} = time that maximum concentration was reached. ^dAUC_{0-t} = area under the plasma concentration–time curve. ^eCL = *in vivo* clearance. ^fV_{ss} = steady state volume of distribution. ^gt_{1/2} = *in vivo* half-life. ^hF = bioavailability. ⁱPredicted values.

position with the most potent being the ethyl substituent of aficamten. Introduction of polar groups led to a reduction in potency as exemplified with methyl ethers **26** and **27** and alcohol **28**. Substitution of the methyl group of **17** with a hydrogen (**21**) led to a reduction of potency. Aficamten (R = ethyl) had the best balance of biochemical potency, micro-

somal stability, and solubility and was selected for advanced characterization.

The final stage of optimization matched the ethyl, cyclopropyl, and isopropyl oxadiazole substituents with a variety of heterocyclic amides (**29–39**) to further improve biochemical potency and to identify compounds with a pharmacokinetic profile in preclinical species that would be

predicted to enable once a day dosing and reaching steady state plasma concentrations within two weeks in humans (Table 6). All compounds within this set demonstrated good biochemical potency and low clearance in human microsomes, but significant differences were observed in rat pharmacokinetics. Aficamten showed a balance of desirable biochemical activity and rat PK and was selected for pharmacokinetic characterization in additional preclinical species. Rat bioavailability was reduced for the free pyrazole analogue of aficamten (32) as well as the oxazole (29 and 31) and diazine (34 and 35) analogues. Tetrazole analogues 30, 36, and 37 also possessed suitable biochemical activity and rat PK, but challenges with the large-scale synthesis of tetrazole amides was expected to lengthen development times for these compounds. The long rat half-life of 36 (18 h) was also not ideal to achieve our human PK objectives.

The PK of aficamten was assessed in multiple preclinical species to enable predictions of human clearance, volume of distribution, and half-life via simple allometry (Table 7). Aficamten was rapidly absorbed after oral administration and showed oral bioavailability values of 41–98% across species. Clearance rates were well below hepatic blood flow in all species, and volumes of distribution were near or above the average for total body water in these species. The resultant human PK predictions (clearance of 2.1 mL/min/kg, half-life of 2.8 days) provided confidence that aficamten could be dosed once a day and reach steady state plasma concentrations within 2 weeks.²³

To assess the therapeutic window and PK/PD relationship relative to a clinical stage compound, the pharmacodynamic dose and exposure response of aficamten and mavacamten were measured by echocardiography in normal, Sprague–Dawley rats. Figure 4 shows the effect of single oral doses of aficamten and mavacamten ranging from 0.5 to 4 mg/kg on cardiac contractility over time. Left ventricular dimensions and fractional shortening (FS) were determined by echocardiography at select time points over a 24-h period. Both aficamten and mavacamten reduced FS dose-related manner, with a maximal reduction in FS between 70 and 80% relative to vehicle treatment observed 1 h after the highest dose of 4 mg/kg. To assess the therapeutic window for aficamten and mavacamten (Figure 5), blood samples were collected concomitant with determination of FS to assess the compound concentrations associated with each FS measurement. The therapeutic window was defined by the IC_{50}/IC_{10} ratio in the echocardiography experiment and corresponded to compound concentrations that reduced FS by 50% (IC_{50}) and 10% (IC_{10}), with a higher IC_{50}/IC_{10} ratio predictive of a greater therapeutic window. Reduction of fractional shortening by greater than 50% has been shown to exhibit a heart failure phenotype in preclinical models and was used to define an excessive reduction in cardiac function.²⁴ The IC_{10} value determined in the echo study represented our estimation of a beneficial *in vivo* effect that was later confirmed in human testing.¹⁰ Aficamten had a shallow concentration–response profile with an IC_{50}/IC_{10} ratio of 9.9 $\times$, whereas mavacamten had a steeper concentration–response profile with an IC_{50}/IC_{10} ratio of 2.8 $\times$.

Cardiac myosin is highly conserved across species,²⁵ and inhibitors of cardiac myosin should behave similarly across species. To strengthen this hypothesis, the therapeutic window and the PK/PD relationship of aficamten was measured in dogs under a similar paradigm as described above for rats. The

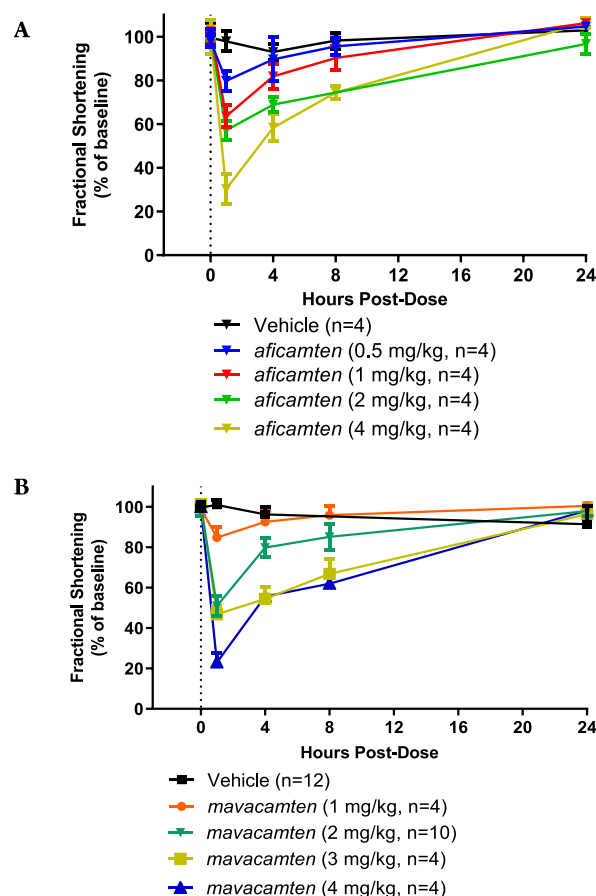
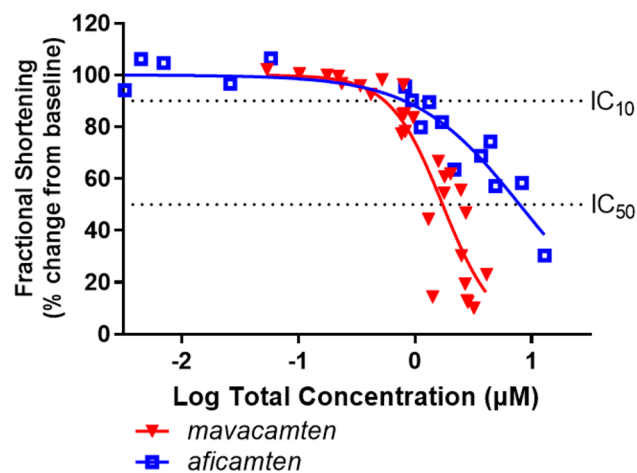


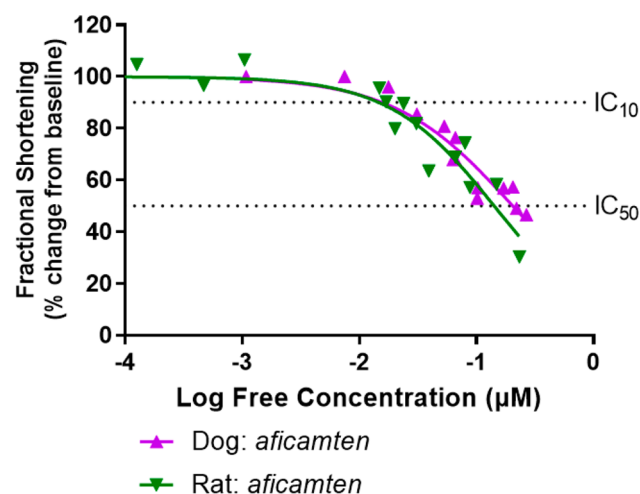
Figure 4. Reduction of FS *in vivo* in Sprague–Dawley rats for (A) aficamten and (B) mavacamten.



	IC_{10} (μM)	IC_{50} (μM)	IC_{50}/IC_{10}
Aficamten	0.8	7.9	9.9
Mavacamten	0.6	1.7	2.8

Figure 5. Reduction of FS *in vivo* in Sprague–Dawley rats vs compound concentration for aficamten and mavacamten.

comparison of the change in FS from baseline vs free drug concentrations in rats and dogs is shown in Figure 6. Free drug



Species	IC _{10-free} (μM)	IC _{50-free} (μM)	IC _{50-free} /IC _{10-free}
rat	0.0144	0.142	9.9
dog	0.0178	0.231	13.0

Figure 6. Reduction of FS *in vivo* in Sprague–Dawley rats and beagle dogs vs free compound concentration for aficamten.

concentrations were used in this comparison due to the differences in plasma protein binding in rats as compared to beagle dogs (1.8 and 25.1% free concentrations, respectively). Aficamten produced similar results in rats and dogs, with a free IC₁₀ (IC_{10-free}) of 0.0144 and 0.0178 μM and an IC_{50-free}/IC_{10-free} ratio of 9.9× and 13.0×, respectively.

To evaluate our third primary objective, the design of a compound with no substantial CYP induction or inhibition to minimize the potential for CYP-induced drug–drug interactions, aficamten was assessed for its direct and time dependent inhibition potential of CYP in human liver microsomes (HLM) and its induction potential in cryopreserved human hepatocytes. Aficamten showed no inhibition against CYPs 1A2, 2B6, 2C8, 2C9, 2C19, 2D6, or 3A4 under the conditions tested (Table 8). Additionally, following a 30

Table 8. Inhibition of CYP Isoenzymes-Mediated Reactions by Aficamten in HLM

CYP	probe CYP reaction	inhibition by aficamten IC ₅₀ (μM)
1A2	phenacetin <i>O</i> -deethylation	>100
2B6	bupropion hydroxylation	>100
2C8	paclitaxel 6α-hydroxylation	>100
2C9	diclofenac 4'-hydroxylation	>100
2C19	<i>S</i> -mephenytoin 4'-hydroxylation	>100
2D6	bufuralol 1'-hydroxylation	>100
3A4	midazolam 1'-hydroxylation	>100
3A4	testosterone 6β-hydroxylation	>100

min preincubation of aficamten in human liver microsomes with and without NADPH, no significant time-dependent inhibition of enzyme activities was found with these same CYP isoforms. The potential of aficamten to induce CYP1A2, CYP2B6, and CYP3A4 was tested in cryopreserved plateable human primary hepatocytes from three separate donors using mRNA levels as the end point. Based on predetermined cutoff values,²⁶ aficamten did not induce CYP2B6 and CYP3A4 in all

three donors at doses up to 50 μM and did not induce CYP1A2 in all three donors at doses up to 25 μM. When tested at 50 μM, two out of the three donors did not show evidence of CYP1A2 induction, but one donor showed a modest induction (2.9-fold induction vs a 2-fold cutoff value) of CYP1A2 at this high concentration.

Aficamten met our optimization program objectives for a novel, next-generation cardiac myosin inhibitor. A comparison of data for aficamten and mavacamten including the three primary optimization objectives is provided in Table 9. The

Table 9. Key Data Comparison of Aficamten and Mavacamten

assay	aficamten	mavacamten
cardiac myofibril IC ₅₀ (μM) ^a	1.26 (1.20–1.33)	0.6 (0.54–0.67)
rat FS IC ₁₀ (μM)	0.8	0.6
rat FS IC ₅₀ /IC ₁₀	9.9	2.8
human <i>t</i> _{1/2} (projected, days)	2.8 ²²	9 ¹¹
human <i>t</i> _{1/2} (actual, days)	3.4 ²⁵	7–9 ^{11–13}
CYP induction EC ₅₀ (μM)	no substantial induction up to 25 μM for 3A4, 2B6, and 1A2	2.2 ± 0.4/3A4 ¹¹ 5.1 ± 0.2/2B6 ¹¹
CYP450 inhibition (IC ₅₀ , μM) ^b	>100	>30 ¹¹
time dependent CYP inhibition	none	NA ^c

^aCytokinetics internal data. ^bMavacamten was tested in CYPs 1A2, 2B6, 2C9, 2C19, 2D6, 3A4, and 3A5; aficamten was tested in 1A2, 2B6, 2C8, 2C9, 2C19, 2D6, and 3A4. ^cNA = data not available.

preclinical pharmacokinetic assessment predicted that aficamten has a projected human half-life appropriate for once a day dosing and would reach steady state within two weeks, thus accomplishing the first objective. The therapeutic window in Sprague–Dawley rats for aficamten and mavacamten, as measured by the IC₅₀/IC₁₀ ratio with higher ratio indicative of a greater therapeutic window in this model, were 9.9× and 2.8×, respectively, achieving the second primary optimization objective. Finally, aficamten showed no substantial CYP induction or inhibition, as measured in human liver microsomes, and human hepatocytes, respectively, completing the third primary optimization objective.

Based on its preclinical profile, aficamten was advanced into human clinical studies, and a phase 1 trial in healthy participants was completed. The study demonstrated that aficamten was safe and well tolerated in healthy participants. No serious adverse events and no clinically meaningful changes in vital signs, ECGs, or laboratory tests were observed. The pharmacokinetics of aficamten were generally dose linear, and steady state appeared evident within 14 days of dosing, in line with the allometric projections made prior to its study in humans. Left ventricular ejection fraction decreased in an exposure dependent manner, and the PK/PD relationship for aficamten observed in humans was similar to that observed preclinically when adjusted for differences in protein binding. Specifically, the shallow exposure–response relationship observed preclinically appeared to translate to humans and thereby may enable flexible dose optimization in humans. The data supported the advancement of aficamten into a Phase 2

clinical trial in patients with obstructive hypertrophic cardiomyopathy that is ongoing.²⁷

CONCLUSION

In conclusion, aficamten is a novel, next-generation cardiac myosin inhibitor that provides a projected human half-life appropriate for once a day dosing, reaching steady state within two weeks, has no substantial CYP induction or inhibition in preclinical assessment and demonstrates a wide therapeutic window *in vivo* with a clear PK/PD relationship. The preclinical data supported progression of aficamten into phase 1 studies where steady state was reached within 14 days of dosing and the wide therapeutic window observed preclinically appeared to translate to humans. Aficamten may provide an advancement to counteracting the hypercontractility of the cardiac sarcomere that appears to underlie pathological hypertrophy, outflow obstruction, and fibrosis in select genetic hypertrophic cardiomyopathies.

EXPERIMENTAL SECTION

General Information. Unless otherwise noted, all solvents and reagents were purchased from commercial vendors and used without further purification. Anhydrous THF, Et₂O, toluene, and CH₂Cl₂ were purchased from standard commercial vendors. All air or moisture sensitive reactions were carried out under an atmosphere of nitrogen in glassware that had been oven- or flame-dried. ¹H NMR and ¹³C NMR spectra were recorded at ambient temperature at 400.13 and 100.62 MHz, respectively, using a Bruker AVANCE 400 spectrometer. ¹H shifts are referenced to the residual protonated solvent signal (δ 2.50 for DMSO-*d*₆, δ 3.31 for DMSO-*d*₄, δ 7.24 for CDCl₃) and ¹³C shifts are referenced to the deuterated solvent signal (δ 39.5 for *d*₆-DMSO and δ 53.8 for CD₂Cl₂). The data are reported as follows: chemical shift in ppm from internal tetramethylsilane on the δ scale, multiplicity (br = broad, s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet), coupling constants (Hz), and integration. Low resolution mass spectrometry data were obtained using an Agilent LC/MSD Quad VL system. Normal phase liquid chromatography was performed using forced flow (flash chromatography) of the indicated solvent system on EM Reagents silica gel (SiO₂, 60 (230–400 mesh) or using a Biotage Horizon MPLC with Biotage KP-Sil silica gel columns. Reverse phase HPLC purification was performed with an Agilent Series 1100 HPLC equipped with a Phenomenex Gemini C18 Column (21.2 mm $\times$ 150 mm, 5 μ m particle size). The typical gradient used for the mobile phase was 20% acetonitrile/water to 90% acetonitrile/water in the presence of 0.1% formic acid over 40 min unless otherwise specified. The purity for compounds 1–39 was judged to be >95% as determined by ¹H NMR and HPLC at 254 nm. All animal experiments described in this manuscript were performed in compliance with Institutional Animal Care and Use Committee (IACUC) guidelines.

***tert*-Butyl (R)-(5-Bromo-2,3-dihydro-1H-inden-1-yl)carbamate.** To a solution of (1R)-5-bromo-2,3-dihydro-1H-inden-1-amine hydrochloride (44.4 g, 178.8 mmol, 1.0 equiv) in CH₂Cl₂ (330 mL) at 0 °C were added TEA (39.8 g, 393.3 mmol, 2.2 equiv) and a solution of (Boc)₂O (42.9 g, 196.3 mmol, 1.1 equiv) in CH₂Cl₂ (120 mL) dropwise over a period of 1 h. The mixture was stirred at rt for 3 h. Water (500 mL) was added, and the mixture was extracted with CH₂Cl₂ (500 mL) twice. The combined organic layers were washed twice with aqueous NH₄Cl solution (500 mL), twice with brine (500 mL), dried over anhydrous Na₂SO₄, and concentrated under reduced pressure to give 57.4 g (92%) of *tert*-butyl (R)-(5-bromo-2,3-dihydro-1H-inden-1-yl)carbamate as a white solid. LRMS (ES): calculated for C₁₄H₁₈BrNO₂, 311.1 Da, measured 256.1 *m/z* [M + H – 56]⁺.

***tert*-Butyl (R)-(5-Cyano-2,3-dihydro-1H-inden-1-yl)carbamate.** To a solution of *tert*-butyl (R)-(5-bromo-2,3-dihydro-1H-inden-1-yl)carbamate (57.4 g, 184 mmol, 1.0 equiv) in a mixture of dioxane (285 mL) and water (285 mL) under a nitrogen atmosphere were

added potassium acetate (36.0 g, 367 mmol, 2.0 equiv), K₄Fe(CN)₆·3H₂O (31.1 g, 73.5 mmol, 0.4 equiv), XPhos (1.3 g, 2.8 mmol, 0.015 equiv), and second generation XPhos precatalyst (2.2 g, 2.8 mmol, 0.015 equiv). The reaction mixture was stirred at 100 °C for 2 h, cooled to rt, and filtered to remove solids. The aqueous layer was extracted with ethyl acetate (500 mL) twice. The combined organic layers were dried over Na₂SO₄, concentrated, and triturated with 10% ethyl acetate/hexanes (300 mL) to give 42.0 g (88%) of *tert*-butyl (R)-(5-cyano-2,3-dihydro-1H-inden-1-yl)carbamate as a light yellow solid. LRMS (ES): calculated for C₁₅H₁₈N₂O₂, 258.1 Da, measured 203.1 *m/z* (M + H – 56).

***tert*-Butyl (R)-(5-(N-Hydroxycarbamimidoyl)-2,3-dihydro-1H-inden-1-yl)carbamate.** To a solution of *tert*-butyl (R)-(5-cyano-2,3-dihydro-1H-inden-1-yl)carbamate (42.2 g, 163.4 mmol, 1 equiv) in ethanol (420 mL) were added hydroxylamine hydrochloride (22.7 g, 326.7 mmol, 2.0 equiv) and TEA (33.1 g, 326.7 mmol, 2.0 equiv). The mixture was stirred at 50 °C for 4 h, concentrated, dissolved in ethyl acetate (1 L), washed with water, dried over Na₂SO₄, and concentrated to give 54.6 g (98%) of *tert*-butyl (R)-(5-(N-hydroxycarbamimidoyl)-2,3-dihydro-1H-inden-1-yl)carbamate as an off-white solid. LRMS (ES): calculated for C₁₅H₂₁N₃O₃, 291.1 Da, measured 236.1 *m/z* (M + H – 56).

***tert*-Butyl (R)-(5-(5-Ethyl-1,2,4-oxadiazol-3-yl)-2,3-dihydro-1H-inden-1-yl)carbamate.** To a solution of *tert*-butyl (R)-(5-(N-hydroxycarbamimidoyl)-2,3-dihydro-1H-inden-1-yl)carbamate (16.0 g, 54.9 mmol, 1.0 equiv) in dioxane (300 mL) was added propanoic anhydride (8.4 g, 64.5 mmol, 1.2 equiv). The mixture was stirred at 105 °C for 8 h, cooled to rt, concentrated, and purified using silica gel chromatography (10% ethyl acetate/petroleum ether) to give 17.5 g (97%) of *tert*-butyl (R)-(5-(5-ethyl-1,2,4-oxadiazol-3-yl)-2,3-dihydro-1H-inden-1-yl)carbamate as a white solid. LRMS (ES): calculated for C₁₈H₂₃N₃O₃, 329.2 Da, measured 274.1 *m/z* (M + H – 56).

(R)-5-(5-Ethyl-1,2,4-oxadiazol-3-yl)-2,3-dihydro-1H-inden-1-amine. To a solution of *tert*-butyl (R)-(5-(5-ethyl-1,2,4-oxadiazol-3-yl)-2,3-dihydro-1H-inden-1-yl)carbamate (17.6 g, 53.4 mmol, 1.0 equiv) in CH₂Cl₂ (120 mL) was added TFA (24 mL). The mixture was stirred at rt overnight and concentrated under reduced pressure. The mixture was then poured into ethanol (50 mL) and water (5 mL), and the pH was adjusted to 12 with an aqueous sodium hydroxide solution (2 N). The mixture was then extracted with CH₂Cl₂ (200 mL) three times. The combined organic layers were dried over anhydrous sodium sulfate and concentrated to give 11.2 g (92%) of (R)-5-(5-ethyl-1,2,4-oxadiazol-3-yl)-2,3-dihydro-1H-inden-1-amine as a brown oil. LRMS (ES): calculated for C₁₃H₁₅N₃O, 229.1 Da, measured 230.1 *m/z* (M + H).

(R)-N-(5-(5-Ethyl-1,2,4-oxadiazol-3-yl)-2,3-dihydro-1H-inden-1-yl)-1-methyl-1H-pyrazole-4-carboxamide (Aficamten). To a solution of 1-methyl-1H-pyrazole-4-carboxylic acid (6.1 g, 48.4 mmol, 1.0 equiv) in DMF (300 mL) were added HOAt (19.8 g, 145.8 mmol, 3.0 equiv), EDC (28.0 g, 146.1 mmol, 3.0 equiv), and DIEA (12.6 g, 96.8 mmol, 2.0 equiv). The reaction mixture was stirred for 15 min and then (R)-5-(5-ethyl-1,2,4-oxadiazol-3-yl)-2,3-dihydro-1H-inden-1-amine (11.2 g, 48.9 mmol, 1.0 equiv) was added. The resulting mixture was stirred at rt for 3 h, diluted with CH₂Cl₂, washed with NH₄Cl solution three times, dried over sodium sulfate, concentrated under reduced pressure, and purified by silica gel chromatography (26% ethyl acetate/petroleum ether) to give a solid product that was triturated with 10% ethyl acetate/petroleum ether to afford 14.5 g (88%) of N-[(1R)-5-(5-ethyl-1,2,4-oxadiazol-3-yl)-2,3-dihydro-1H-inden-1-yl]-1-methyl-1H-pyrazole-4-carboxamide as a white solid. Spectral data for aficamten: ¹H NMR (300 MHz, DMSO-*d*₆) δ 8.41 (1H, d, *J* = 8.4 Hz), 8.16 (1H, s), 7.91–7.79 (3H, m), 7.34 (1H, d, *J* = 7.9 Hz), 5.53 (1H, q, *J* = 8.3 Hz), 3.84 (3H, s), 3.13–2.81 (4H, m), 2.44 (1H, dd, *J* = 7.9, 4.7 Hz), 1.95 (1H, m), 1.33 (3H, t, *J* = 7.5 Hz). ¹³C NMR (101 MHz, DMSO) δ 180.98, 167.59, 161.65, 148.01, 144.00, 138.51, 132.11, 125.50, 125.47, 124.63, 122.98, 118.23, 53.33, 40.10, 39.89, 39.68, 39.47, 39.26, 39.16, 39.05, 38.85, 38.69, 32.80, 29.58, 19.56, 10.38. HRMS (ES): calculated for C₁₈H₁₉N₅O₂, 338.1612 Da [M + H]⁺, measured 338.1597 *m/z* [M + H]⁺.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jmedchem.1c01290>.

Experimental procedures and analytical data for compounds 1–39 and description and experimental procedures for biological and DMPK assays (PDF)

Off-target data for compounds 2, 14, and 37 (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Chihyuan Chuang – Cytokinetics, Inc., South San Francisco, California 94080, United States; orcid.org/0000-0003-1581-2303; Phone: 650-624-2914; Email: gchuang@cytokinetics.com; Fax: 650-624-3010

Bradley P. Morgan – Cytokinetics, Inc., South San Francisco, California 94080, United States; Phone: 650-624-3000; Email: bmorgan@cytokinetics.com; Fax: 650-624-3010

Authors

Scott Collibee – Cytokinetics, Inc., South San Francisco, California 94080, United States; orcid.org/0000-0003-3513-4060

Luke Ashcraft – Cytokinetics, Inc., South San Francisco, California 94080, United States

Wenyue Wang – Cytokinetics, Inc., South San Francisco, California 94080, United States

Mark Vander Wal – Cytokinetics, Inc., South San Francisco, California 94080, United States; Present Address: Institute for Neurodegenerative Diseases (IND), and Department of Neurology, Weill Institute for Neurosciences, University of California San Francisco, San Francisco, California 94158, United States

Xiaolin Wang – Cytokinetics, Inc., South San Francisco, California 94080, United States; Present Address: Pharmaron Beijing Limited Co., 6 Taihe Road, BDA, Beijing, 100176 P.R. China

Darren T. Hwee – Cytokinetics, Inc., South San Francisco, California 94080, United States

Yangsong Wu – Cytokinetics, Inc., South San Francisco, California 94080, United States

Jingying Wang – Cytokinetics, Inc., South San Francisco, California 94080, United States; Present Address: Amgen, 1120 Veterans Blvd, South San Francisco, California 94030, United States

Eva R. Chin – Cytokinetics, Inc., South San Francisco, California 94080, United States; Present Address: NMD Pharma, Palle Juul-Jensens Boulevard 82, 8200 Aarhus N, Denmark

Peadar Cremin – Cytokinetics, Inc., South San Francisco, California 94080, United States; Present Address: Kodiak Sciences, 2631 Hanover Street, Palo Alto, California 94304, United States

Jeanette Zamora – Cytokinetics, Inc., South San Francisco, California 94080, United States

James Hartman – Cytokinetics, Inc., South San Francisco, California 94080, United States

Julia Schaletzky – Cytokinetics, Inc., South San Francisco, California 94080, United States; Present Address: The Center for Emerging and Neglected Diseases, University of California Berkeley, 344A Li Ka Shing, Berkeley, California 94720, United States

Eddie Wehri – Cytokinetics, Inc., South San Francisco, California 94080, United States; Present Address: The Center for Emerging and Neglected Diseases, University of California Berkeley, 344A Li Ka Shing, Berkeley, California 94720, United States

Laura A. Robertson – Cytokinetics, Inc., South San Francisco, California 94080, United States

Fady I. Malik – Cytokinetics, Inc., South San Francisco, California 94080, United States

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acs.jmedchem.1c01290>

Notes

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■ ABBREVIATIONS

Boc, *tert*-butoxycarbonyl; Cbz, benzyloxycarbonyl; CDMF Ca₇₅, cardiac myofibril ATPase assay with Ca concentration @ 75% of the maximum contraction; Cl, clearance; Cl_{int}, intrinsic clearance; cLogP, calculated LogP; CMI, cardiac myosin inhibitor; CSI, cardiac sarcomere inhibitor; CYP, cytochrome P-450; DDIs, pharmacokinetic drug–drug interactions; F, bioavailability; FS, fractional shortening; HCM, hypertrophic cardiomyopathy; HLM, human liver microsome; HTS, high-throughput screen; IC₁₅, 15% inhibitory concentration; IC₅₀, 50% inhibitory concentration; LE, ligand efficiency; LVEF, left ventricular ejection fraction; oHCM, obstructive HCM; PD, pharmacodynamic; PK, pharmacokinetics; PPB, plasma protein binding; pVo₂, peak oxygen consumption; qd, once daily dosing interval; t_{1/2}, half-life

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